
Consciousness in Artificial Intelligence: Insights from the Science of Consciousness

Patrick Butlin*	Robert Long*	Eric Elmoznino
Yoshua Bengio	Jonathan Birch	Axel Constant
George Deane	Stephen M. Fleming	Chris Frith
Xu Ji	Ryota Kanai	Colin Klein
Grace Lindsay	Matthias Michel	Liad Mudrik
Megan A. K. Peters	Eric Schwitzgebel	Jonathan Simon
	Rufin VanRullen	

Abstract

Whether current or near-term AI systems could be conscious is a topic of scientific interest and increasing public concern. This report argues for, and exemplifies, a rigorous and empirically grounded approach to AI consciousness: assessing existing AI systems in detail, in light of our best-supported neuroscientific theories of consciousness. We survey several prominent scientific theories of consciousness, including recurrent processing theory, global workspace theory, higher-order theories, predictive processing, and attention schema theory. From these theories we derive "indicator properties" of consciousness, elucidated in computational terms that allow us to assess AI systems for these properties. We use these indicator properties to assess several recent AI systems, and we discuss how future systems might implement them. Our analysis suggests that no current AI systems are conscious, but also suggests that there are no obvious technical barriers to building AI systems which satisfy these indicators.¹

* Joint first authors and corresponding authors (patrickbutlin@gmail.com, rgblong@gmail.com)

¹ A previous version of this sentence read "...but also shows that there are no obvious barriers to building conscious AI systems." We have amended it to better reflect the messaging of the report: that satisfying these indicators may be feasible. But satisfying the indicators would not mean that such an AI system would definitely be conscious.

Authors

Patrick Butlin*, Future of Humanity Institute, University of Oxford

Robert Long*, Center for AI Safety

Eric Elmoznino, University of Montreal and MILA - Quebec AI Institute

Yoshua Bengio, University of Montreal and MILA - Quebec AI Institute

Jonathan Birch, Centre for Philosophy of Natural and Social Science, London School of Economics and Political Science

Axel Constant, School of Engineering and Informatics, The University of Sussex and Centre de Recherche en Éthique, University of Montreal

George Deane, Department of Philosophy, University of Montreal

Stephen M. Fleming, Department of Experimental Psychology and Wellcome Centre for Human Neuroimaging, University College London

Chris Frith, Wellcome Centre for Human Neuroimaging, University College London and Institute of Philosophy, University of London

Xu Ji, University of Montreal and MILA - Quebec AI Institute

Ryota Kanai, Araya, Inc.

Colin Klein, School of Philosophy, The Australian National University

Grace Lindsay, Department of Psychology and Center for Data Science, New York University

Matthias Michel, Center for Mind, Brain and Consciousness, New York University

Liad Mudrik, School of Psychological Sciences and Sagol School of Neuroscience, Tel-Aviv University and CIFAR Program in Brain, Mind and Consciousness

Megan A. K. Peters, Department of Cognitive Sciences, University of California, Irvine and CIFAR Program in Brain, Mind and Consciousness

Eric Schwitzgebel, Department of Philosophy, University of California, Riverside

Jonathan Simon, Department of Philosophy, University of Montreal

Rufin VanRullen, Centre de Recherche Cerveau et Cognition, CNRS, Université de Toulouse

Executive Summary

The question of whether AI systems could be conscious is increasingly pressing. Progress in AI has been startlingly rapid, and leading researchers are taking inspiration from functions associated with consciousness in human brains in efforts to further enhance AI capabilities. Meanwhile, the rise of AI systems that can convincingly imitate human conversation will likely cause many people to believe that the systems they interact with are conscious. In this report, we argue that consciousness in AI is best assessed by drawing on neuroscientific theories of consciousness. We describe prominent theories of this kind and investigate their implications for AI.

We take our principal contributions in this report to be:

1. Showing that the assessment of consciousness in AI is scientifically tractable because consciousness can be studied scientifically and findings from this research are applicable to AI;
2. Proposing a rubric for assessing consciousness in AI in the form of a list of indicator properties derived from scientific theories;
3. Providing initial evidence that many of the indicator properties can be implemented in AI systems using current techniques, although no current system appears to be a strong candidate for consciousness.

The rubric we propose is provisional, in that we expect the list of indicator properties we would include to change as research continues.

Our method for studying consciousness in AI has three main tenets. First, we adopt *computational functionalism*, the thesis that performing computations of the right kind is necessary and sufficient for consciousness, as a working hypothesis. This thesis is a mainstream—although disputed—position in philosophy of mind. We adopt this hypothesis for pragmatic reasons: unlike rival views, it entails that consciousness in AI is possible in principle and that studying the workings of AI systems is relevant to determining whether they are likely to be conscious. This means that it is productive to consider what the implications for AI consciousness would be if computational functionalism were true. Second, we claim that neuroscientific theories of consciousness enjoy meaningful empirical support and can help us to assess consciousness in AI. These theories aim to identify functions that are necessary and sufficient for consciousness in humans, and computational functionalism implies that similar functions would be sufficient for consciousness in AI. Third, we argue that a *theory-heavy approach* is most suitable for investigating consciousness in AI. This involves investigating whether AI systems perform functions similar to those that scientific theories associate with consciousness, then assigning credences based on (a) the similarity of the functions, (b) the strength of the evidence for the theories in question, and (c) one's credence in computational functionalism. The main alternative to this approach is to use behavioural tests for consciousness, but this method is unreliable because AI systems can be trained to mimic human behaviours while working in very different ways.

Various theories are currently live candidates in the science of consciousness, so we do not endorse any one theory here. Instead, we derive a list of *indicator properties* from a survey of theories of consciousness. Each of these indicator properties is said to be necessary for consciousness

by one or more theories, and some subsets are said to be jointly sufficient. Our claim, however, is that AI systems which possess more of the indicator properties are more likely to be conscious. To judge whether an existing or proposed AI system is a serious candidate for consciousness, one should assess whether it has or would have these properties.

The scientific theories we discuss include recurrent processing theory, global workspace theory, computational higher-order theories, and others. We do not consider integrated information theory, because it is not compatible with computational functionalism. We also consider the possibility that agency and embodiment are indicator properties, although these must be understood in terms of the computational features that they imply. This yields the following list of indicator properties:

Recurrent processing theory
RPT-1: Input modules using algorithmic recurrence
RPT-2: Input modules generating organised, integrated perceptual representations
Global workspace theory
GWT-1: Multiple specialised systems capable of operating in parallel (modules)
GWT-2: Limited capacity workspace, entailing a bottleneck in information flow and a selective attention mechanism
GWT-3: Global broadcast: availability of information in the workspace to all modules
GWT-4: State-dependent attention, giving rise to the capacity to use the workspace to query modules in succession to perform complex tasks
Computational higher-order theories
HOT-1: Generative, top-down or noisy perception modules
HOT-2: Metacognitive monitoring distinguishing reliable perceptual representations from noise
HOT-3: Agency guided by a general belief-formation and action selection system, and a strong disposition to update beliefs in accordance with the outputs of metacognitive monitoring
HOT-4: Sparse and smooth coding generating a “quality space”
Attention schema theory
AST-1: A predictive model representing and enabling control over the current state of attention
Predictive processing
PP-1: Input modules using predictive coding
Agency and embodiment
AE-1: Agency: Learning from feedback and selecting outputs so as to pursue goals, especially where this involves flexible responsiveness to competing goals
AE-2: Embodiment: Modeling output-input contingencies, including some systematic effects, and using this model in perception or control

Table 1: Indicator Properties

We outline the theories on which these properties are based and describe the evidence and arguments that support them in section 2 of the report, as well as explain the formulations used in the table.

Having formulated this list of indicator properties, in section 3.1 we discuss how AI systems could be constructed, or have been constructed, with each of the indicator properties. In most cases, standard machine learning methods could be used to build systems that possess individual properties from this list, although experimentation would be needed to learn how to build and train functional systems which combine multiple properties. There are some properties in the list which are already clearly met by existing AI systems (such as RPT-1, algorithmic recurrence), and others where this is arguably the case (such as the first part of AE-1, agency). Researchers have also experimented with systems designed to implement particular theories of consciousness, including global workspace theory and attention schema theory.

In section 3.2, we consider whether some specific existing AI systems possess the indicator properties. These include Transformer-based large language models and the Perceiver architecture, which we analyse with respect to the global workspace theory. We also analyse DeepMind’s Adaptive Agent, which is a reinforcement learning agent operating in a 3D virtual environment; a system trained to perform tasks by controlling a virtual rodent body; and PaLM-E, which has been described as an “embodied multimodal language model”. We use these three systems as case studies to illustrate the indicator properties concerning agency and embodiment. This work does not suggest that any existing AI system is a strong candidate for consciousness.

This report is far from the final word on these topics. We strongly recommend support for further research on the science of consciousness and its application to AI. We also recommend urgent consideration of the moral and social risks of building conscious AI systems, a topic which we do not address in this report. The evidence we consider suggests that, if computational functionalism is true, conscious AI systems could realistically be built in the near term.

fulfillment from artificial systems may be more prone to attributing consciousness to them. Assigning human-like traits to AI can help people cope with difficult emotions or situations, such as feeling more comfortable confiding in an AI that appears empathetic or understanding (Turkle 2011). The use of artificial systems such as Replika chatbots as sources of social interaction is already evident.

The recent rapid progress in language model capabilities is particularly likely to drive over-attribution of consciousness, as the case of Blake Lemoine arguably illustrates (Lemoine 2022). Modern LLMs can convincingly imitate human discourse, making it difficult to resist the impression that one is interacting with a conscious agent, especially if the model is prompted to play the role of a person in a conversation (Shanahan et al. 2023).

In addition to the risk of misallocation of resources which we mentioned above, over-attributing consciousness to AI systems creates risks of at least three other kinds. First, if consciousness is sometimes attributed to AI systems on weak grounds, these attributions may undermine better-evidenced claims of consciousness in AI. Observers of debates on this topic may recognise that some attributions are weak, and assume that all are. This effect could be particularly powerful if there are also reasonable concerns that attributions of consciousness to AI are distracting us from addressing other pressing problems. Second, if we judge that a class of AI systems are conscious, this should lead us to treat them differently—training them in different ways, for instance. In principle, this could conflict with work to ensure that AI systems are developed in ways that benefit society. And third, overattribution could interfere with valuable human relationships, as individuals increasingly turn to artificial agents for social interaction and emotional support. People who do this could also be particularly vulnerable to manipulation and exploitation.

Whatever one’s views about the relative importance of these various risks, they amount to a powerful case for research on the prospect of consciousness in AI. If the development of AI is to continue, this research will be crucial: it is risky to judge without good grounds either that no AI systems can be conscious, or that those of a particular class are conscious.

4.2 Consciousness and Capabilities

In the popular imagination, consciousness is associated with free will, intelligence and the tendency to feel human emotions, including empathy, love, guilt, anger and jealousy. So our suggestion that conscious AI may be possible in the near-term might be taken to imply that we will soon have AI systems akin to the very human-like AIs depicted in science fiction. Whether this in fact follows depends on the relationships between consciousness and other cognitive traits and capacities. Furthermore, conscious AI systems are more likely to be built if consciousness is (or is expected to be) associated with valuable capabilities in AI. So in this subsection, we briefly consider how consciousness might be related to differences in AI systems’ behaviour and capabilities.

One possible argument for the view that we are likely to build conscious AI is that consciousness is associated with greater capabilities in animals, so we will build conscious AI systems in the course of pursuing more capable AI. It is true that scientific theories of consciousness typically claim that conscious experience arises in connection with adaptive traits, selected for the contributions they make to cognitive performance in humans and some other animals. For example, Baars

(1988, 1997) claims that the global workspace “optimizes the trade-off between organization and flexibility” (1988, p. 348), a trade-off that advanced AI systems must also presumably manage.

A weakness of this argument is that human and animal minds are not necessarily a good guide to the connection between consciousness and capabilities in artificial systems. This is because the “design” of animal minds is explained not only by the adaptive value of our capabilities but also by the constraints under which we evolved, which include limits on the quantity and form of data from which we can learn; limits on the amount of energy available to power our minds; and the forms of our ancestors’ minds and the availability of relevant mutations. The space of possible designs for AI systems is different from the space of possible mind designs in biology (Summerfield 2022). So we may well find ways to build high-performing AI systems which are not conscious.

However, some influential AI researchers are currently pursuing projects that aim to increase AI capabilities by building systems that are more likely to be conscious. We have already discussed the ongoing work by Bengio and colleagues which uses ideas from GWT. Goyal and Bengio (2022) write that:

Our aim is to take inspiration from (and further develop) research into the cognitive science of conscious processing, to deliver greatly enhanced AI, with abilities observed in humans thanks to high-level reasoning.

The aim here is to build AI systems that implement at least some of the features underlying consciousness in humans, specifically in order to enhance capabilities. Similarly, LeCun’s proposed architecture for autonomous intelligence has features such as a world model for planning and a “configurator”, described as an element that “takes inputs from all other modules and configures [the other modules] for the task at hand” (LeCun 2022, p. 6). LeCun also suggests that the existence of elements like these in the human brain may be responsible for what he calls “the illusion of consciousness”. These are examples of a widespread and long-established practice in AI of drawing on insights from the cognitive sciences (Hassabis et al. 2017, Zador et al. 2022). So regardless of whether building in consciousness-associated features is the only possible route to greater capabilities, it is one that we are likely to take.

Turning now to the issue of how conscious artificial systems are likely to behave, there are both conceptual and empirical reasons to doubt that being conscious implies having human-like motives or emotions. Conceptually, to be conscious is simply to have subjective experiences. In principle, there could be conscious subjects who had experiences very unlike ours. In particular, there is no apparent conceptual incoherence in the idea of motivationally neutral conscious experiences which lack valence or affect—and even if a conscious subject does have valenced or affective states, these may be triggered by different circumstances from those which trigger our emotions, and motivate them to behave in different ways.

Empirically, the theories of consciousness we have discussed do not generally claim that consciousness implies human-like motives or emotions. One exception is the view that consciousness is only possible in the context of self-maintenance, which we discussed in section 2.4.5, since this presumably entails not only that conscious beings engage in self-maintenance, but that they are motivated to do so. But this view is an outlier. Some theories, such as GWT and PRM, suggest that conscious subjects must be agents and perhaps that their desires are likely to enter into

conscious experience—in GWT, the workspace maintains representations that provide important context for processing in the modules, and current desires might fit this description. But these theories are neutral about what motivates conscious subjects. AST is a somewhat more complicated case: Graziano (2013) argues that the use of an attention schema is important for social cognition because it allows one to model attention either in oneself or in others. He further claims that this is a reason to build conscious AI because AI systems with attention schemas will be capable of empathy (Graziano 2017). But the claim here seems to be that implementing AST is necessary for attributing conscious states to others, and, therefore, for being motivated by empathy to help them, not that consciousness is sufficient for empathetic motives.

Many current concerns about the impacts of AI do not turn on whether AI systems might be conscious. For example, the concern that AI systems trained on data that reflects the current structure of society could perpetuate or exacerbate injustices does not turn on AI consciousness. Nor does the concern that AI could enable various forms of repression, or that AI systems could replace human workers in most jobs (although the economic value of replacing human workers may motivate capabilities research which leads to conscious AI, as we have noted). Perhaps most surprisingly, arguments that AI could pose an existential risk to humanity do not assume consciousness. A typical argument for this conclusion relies on the premises that (i) we will build AI systems that are very highly capable of making and executing plans to achieve goals and (ii) if we give these systems goals that are not well chosen then the methods that they find to pursue them may be extremely harmful (see e.g. Hilton 2022). Neither these premises nor the ways in which they are typically elaborated and defended rely on AI systems being conscious.

4.3 Recommendations

Several authors have made important practical recommendations concerning the possibility of AI consciousness. These include: Bryson’s (2010) case that conscious AI should be avoided; Metzinger’s (2021) call for a moratorium on work that could lead to conscious AI; Graziano’s (2017) arguments in favour of conscious AI; Schwitzgebel and Garza’s (2020) argument that we should only build particular AI systems if we can either be confident that they will be conscious, or be confident that they will not be; and the detailed recommendations of Bostrom and Shulman (2022). We have not assessed the arguments for these recommendations and do not consider them further here.

However, we do recommend support for research on the science of consciousness and its application to AI (as recommended in the AMCS open letter on this subject; AMCS 2023), and the use of the theory-heavy method in assessing consciousness in AI. One way that we can make progress in learning which AI systems are likely to be conscious is by developing and testing scientific theories of consciousness; theoretical refinements within the existing paradigms are valuable, as well as attempts to test predictions of competing theories, as in the ongoing Cogitate adversarial collaboration (Melloni et al. 2021). A second is by undertaking careful empirical research to extend theories of consciousness to non-human animals. This will help us to establish a more general account of the correlates of consciousness, based on evidence from a wider range of cases (Andrews & Birch 2023). And a third is research that refines theories of consciousness specifically in the context of AI. Research of this kind may involve theorising about AI implementations of mecha-

nisms implicated in theories of consciousness; building such systems and testing their capacities; identifying ambiguities in existing theories; and developing and defending more precise formulations of theories, so that their implications for AI are clearer. Integrating work of this kind with continued empirical research on human and animal consciousness can be expected to be especially productive.

Two other lines of research may be important. One is research specifically on valenced and affective consciousness; if conscious experience of this kind is especially morally important, as seems to be the case, then we have a pressing need for well-founded computational theories which can be applied to AI. The other is efforts to develop better behavioural tests for consciousness in AI. Although our view is that the theory-heavy approach is currently the most promising, it may be possible that behavioural tests could be developed which are difficult to game and based on compelling rationales, perhaps informed by theories. If such tests can be developed, they may have practical advantages over theory-heavy assessments.

As we learn more about consciousness, and gain better tools to assess whether it may be present in AI systems, we must also use these tools effectively. That means applying the theory-heavy method as we develop new kinds of AI systems, both prospectively—in assessing the likelihood of consciousness in planned systems, before we build them—and retrospectively. In retrospective evaluation, methods for mechanistic interpretability may be important, since it is possible that systems that do not have potentially consciousness-supporting features built in could acquire them in the course of training. It also means that scientific theories should be central to consciousness-related AI policy and regulation.

Box 4: Open questions about consciousness in AI

This report is not the last word on AI consciousness. Far from it: one of our major aims is to spur further research. Further progress in the neuroscience of consciousness will contribute to our understanding of the conditions for consciousness in AI, and we have suggested that research on consciousness in non-human animals may be particularly valuable. Here, we highlight research questions that are relevant for understanding AI consciousness in particular.

Refining and extending our approach

While following the same basic approach as this report, further research could:

- Examine other plausible theories of consciousness, not considered in this report, and use them to derive further indicators of consciousness;
- Refine or revise the indicators which we have derived from the theories considered here;
- Conduct assessments of other AI systems, or investigate different ways in which the indicators could be implemented.

Furthermore, our approach could be extended by developing a formal evaluation procedure for consciousness that could be applied to AI systems, although it is questionable whether

this would be justified at present, in the context of significant uncertainty about computational functionalism and about particular scientific theories.

Computational functionalism and rival views

Determining whether consciousness is possible on conventional computer hardware is a difficult problem, but progress on it would be particularly valuable, and philosophical research could contribute to such progress. For example, sceptics of computational functionalism have noted that living organisms are not only self-maintaining homeostatic systems but are made up of cells that themselves engage in active self-maintenance (e.g. Seth 2021, Aru et al. 2023); further work could clarify why this might matter for consciousness. Research might also examine whether there are features of standard computers which might be inconsistent with consciousness, but would not be present in unconventional (e.g. neuromorphic) silicon hardware. A further topic for philosophical research is the individuation of AI systems, given that they can be copied, distributed, called in multiple places at once, and so forth.

Valence and phenomenal character in AI

We have set aside the question of what *kinds* of experiences conscious AI systems might have. In principle, these could be quite different from human experiences. An important issue is what it would take for AI systems to be capable of having valenced conscious experiences—that is, ones that feel good or bad—because these might have special moral significance. Research on computational theories of valence would help to address this issue.

AI interpretability research

A significant obstacle for the theory-heavy approach is our limited understanding of the inner workings of complex deep learning systems. AI interpretability research seeks to illuminate and explain these workings, and can, therefore, contribute to many forms of research on AI, including research on consciousness.

Behavioural tests and introspection

Although we have stressed the limitations of behavioural tests for consciousness in AI, we are open to the possibility that compelling tests could be developed, so research towards this end may be valuable. Behavioural tests need not be “theory-neutral”, but could be based on specific scientific theories. Another possible approach is to develop AI systems with reliable introspective capacities that could allow them to meaningfully report on their own consciousness or lack thereof (Long forthcoming).

The ethics of research on AI consciousness

On balance, we believe that research to better understand the mechanisms which might underlie consciousness in AI is beneficial. However, research on this topic runs the risk of building (or enabling others to build) a conscious AI system, which should not be done lightly. Mitigating this kind of risk should be carefully weighed against the value of better understanding consciousness in AI.